ARTICLE | RESEARCH PREPRINT

Statistical Distributions and Continued-Fraction Staging for Long-Term Stroke Repair with Repetitive Transcranial Magnetic Stimulation

Cartik Sharma, NeuroMorph Computational Neuromodulation Group | September 2026

Abstract. We present a hypothesis-generating computational framework for staging repetitive transcranial magnetic stimulation (rTMS) after stroke. The model combines distributional uncertainty around a motor-recovery trajectory with finite-state stage transitions and rational continued-fraction approximants for phase scheduling. Outputs are simulations, not clinical evidence or treatment instructions.

1. Model and finite staging

A bounded Fugl-Meyer motor score is represented by a normal distribution whose mean follows saturating recovery. A prime-modulus state tracks discrete stage transitions, while convergents of a target synchronization ratio provide reproducible scheduling fractions.

$$Y_t \sim \mathcal{N}(\mu_t, \sigma_t^2), \quad \mu_t = B + (A - B)(1 - e^{-kt})$$

$$s_{t+1} = (s_t + \lfloor \gamma r_t \rfloor) \bmod p, \quad s_t \in \mathbb{F}_p$$

$$\rho = [a_0; a_1, \ldots, a_n] \approx p_n / q_n$$

2. Simulated trajectories

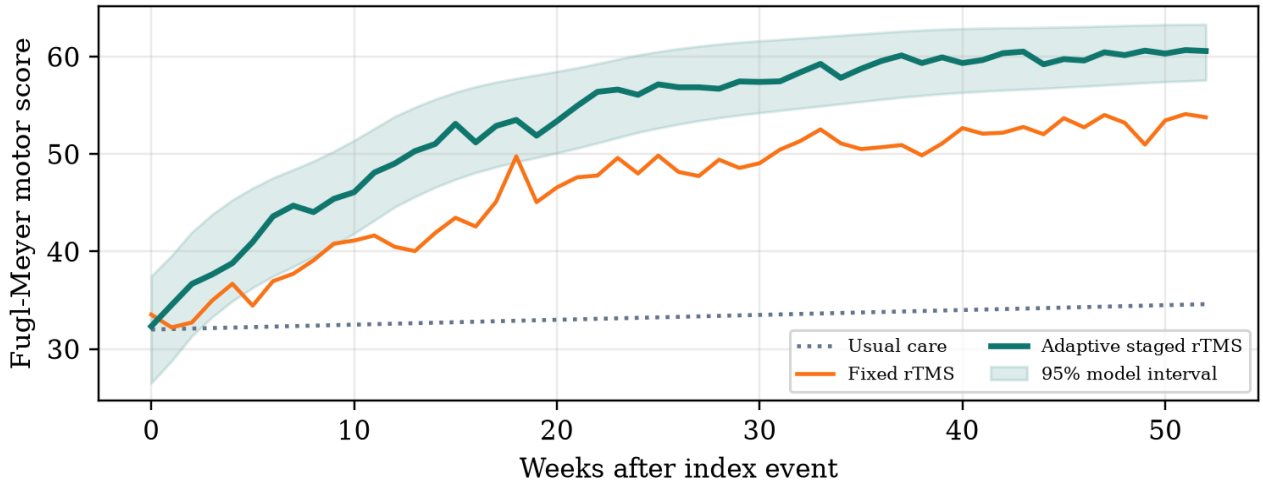


Figure 1 | Simulated long-term trajectories with a distributional uncertainty band. Parameters are illustrative and require prospective validation.

3. Optimal staging protocol

Stage	Weeks	Hz	% MSO	Sessions/wk	CF sync
stabilize	2	1.0	80	3	1/1
prime	4	10.0	90	5	2/1
potentiate	6	20.0	90	5	3/2
consolidate	12	10.0	85	3	5/3
maintain	29	10.0	80	1	8/5

4. Limitations and translation

This preprint is a computational design study. Statistical distributions encode model uncertainty rather than patient-specific confidence intervals. Continued fractions provide numerical scheduling structure, not evidence of biological optimality. Any clinical protocol must be established by qualified clinicians under appropriate ethics, safety, and prospective trial oversight.

Reproducibility: deterministic seed 20260924; default horizon 52 weeks; prime modulus p = 13; six continued-fraction terms.